
SET

Cardinality of a set

Cardinality of a set is the number of elements in that set.

$$A = \{\text{Age, Income, BMI, Height}\}$$

$$|A| = 4$$

$\therefore$ The cardinality of A is 4.

Types of set

Finite set

Finite set is a collection of finite, well-defined elements. The finite set is countable and the set contains a finite number of elements.

ML algorithms in a course.

$$A = \{1, 2, 3, 4, 5\}$$

$$|A| = 5$$

$\therefore$ A is a finite set.

Infinite set

Infinite set is a collection of infinite number of elements. Number of elements in infinite set is infinite.

Possible real-valued predictions

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$$

$\therefore \mathbb{N}$ is an infinite set.

Null set

Null set is a set that contains no elements. Number of elements in a null set is zero. It is also known as empty set.

Dataset records satisfying an impossible condition

$$A = \emptyset$$

$$|A| = 0$$

$\therefore$ A is a null set.

Equal set

Two sets are said to be equal if they contain the same elements, regardless of the order in which they are listed.

Two feature sets containing the same features

$$A = \{1, 2, 3\}, \quad B = \{3, 1, 2\}$$

$$\therefore A = B$$

Equivalent set

Two or more sets are said to be equivalent if they have the same number of elements, regardless of what the elements are. Cardinality of equivalent set are equal. If set A and set B are equivalent set then the number of elements of A will be equal to number of elements of B . ie, $n(A)=n(B)$

Two datasets with the same number of features

$$A = \{1, 2, 3\}, \quad B = \{a, b, c\}$$

$$|A| = 3, \quad |B| = 3$$

$$\therefore A \sim B$$

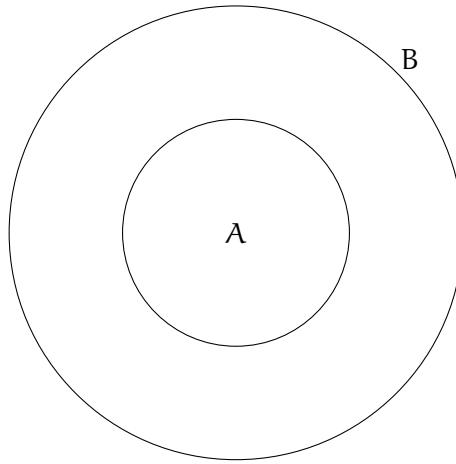
Hence, A and B are equivalent sets.

Subset

Set A is said to be a subset of B if every elements of A is also an element of B.

Selected features from a dataset

$$A \subseteq B \iff \forall x (x \in A \Rightarrow x \in B)$$



$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}$$

$$A \subseteq B$$

$\therefore$ A is a subset of B.

Proper subset

A set A is a proper subset of B if every element of A is in B, but A not equal to B.

Selected features excluding at least one feature

$$A \subset B \iff A \subseteq B \text{ and } A \neq B$$

$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}$$

$$A \subsetneq B$$

$\therefore$ A is a proper subset of B.

Superset

A set B is said to be superset of A if every element of A is also an element of B.

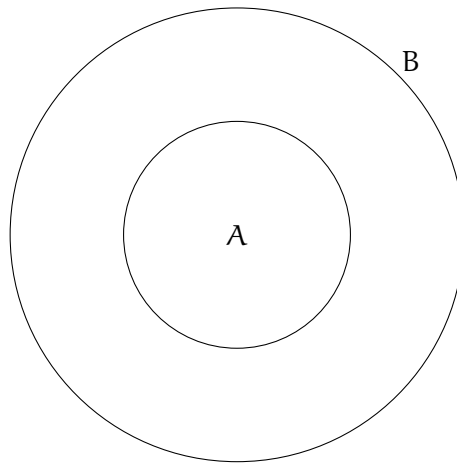
Complete feature set

$$B \supseteq A \iff \forall x (x \in A \Rightarrow x \in B)$$

$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}$$

$$B \supseteq A$$

$\therefore$ B is a superset of A.



Universal set

The universal set is the set that contains all the elements under consideration.

All features available in a dataset

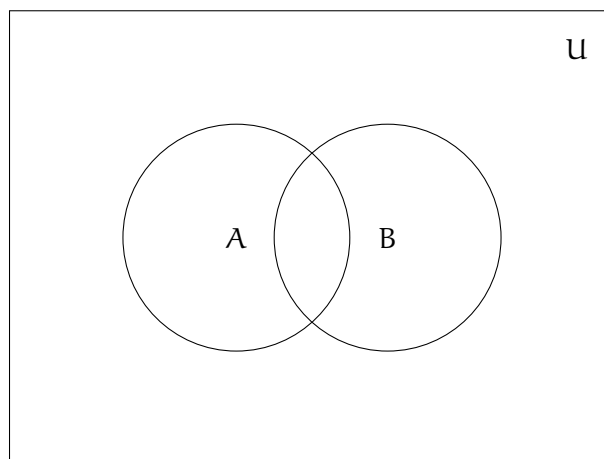
$$U = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 2, 3\}, \quad B = \{4, 5\}$$

$$A \subseteq U \quad \text{and} \quad B \subseteq U$$

$\therefore U$ is the universal set.

tikz



$$A \subseteq U \quad \text{and} \quad B \subseteq U$$

The rectangle is U , and the circles A and B are subsets of U .

Power set

The power set of a set A is the set containing all possible subsets of A , including the empty set and A itself.

All possible feature combinations

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}$$

$$A = \{1, 2\}$$

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

Singleton set

It is a set with only one element. Cardinality of a singleton set is one. If A is a singleton set then $n(A)=1$.

Dataset with only one class

$$A = \{5\}$$

$$|A| = 1$$

$\therefore$ A is a singleton set.

Countable set

A set is called countable if its elements can be put into a one-to-one correspondence with the natural numbers $\mathbb{N}$, either for a finite set or an infinite set.

Set of training epochs

A is countable $\iff \exists f : \mathbb{N} \rightarrow A$ such that f is bijective

$$A = \mathbb{N} = \{1, 2, 3, 4, \dots\}$$

$$A = \{a_1, a_2, a_3, \dots\}$$

$\therefore$ A is a countable set.

Operations on set

Union

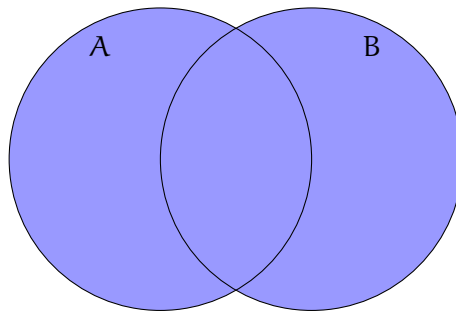
The union of two sets A and B contains all elements that are in A, in B, or in both.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

$$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\}$$

$$A \cup B = \{1, 2, 3, 4, 5\}$$

$\therefore$ $A \cup B$ is the union of A and B.



Intersection

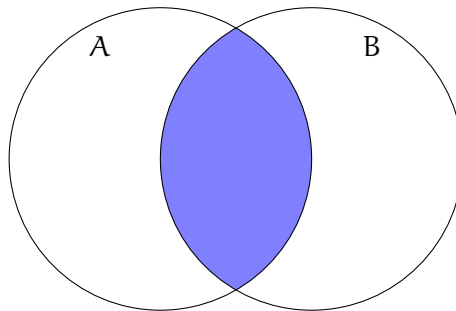
The intersection of two sets A and B contains the elements that are common to both sets.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

$$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\}$$

$$A \cap B = \{3\}$$

$\therefore A \cap B$ is the intersection of A and B.



Set difference

The difference of two sets A and B, written $A - B$, contains the elements that are in A but not in B.

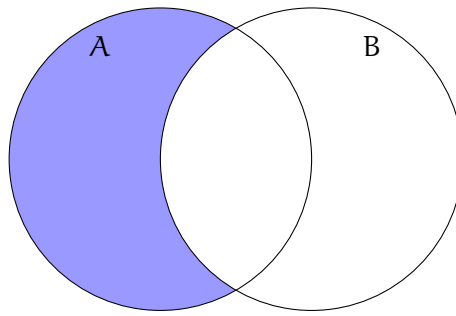
$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}$$

$$A - B = \{1, 2\}$$

$\therefore A - B$ is the difference of A and B.

$$B - A = \{5, 6\}$$



Symmetric difference

The symmetric difference of two sets A and B contains the elements that are in either A or B, but not in both.

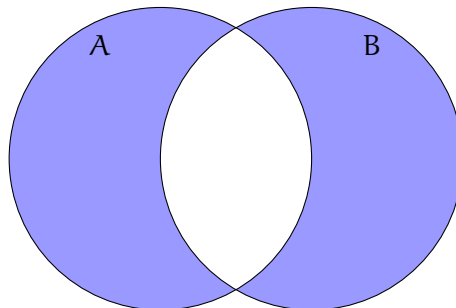
$$A \Delta B = (A - B) \cup (B - A)$$

$$A \Delta B = (A \cup B) - (A \cap B)$$

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}$$

$$A \Delta B = \{1, 2, 5, 6\}$$

$\therefore A \Delta B$ is the symmetric difference of A and B.



Complement

The complement of a set A contains all the elements in the universal set U that are not in A.

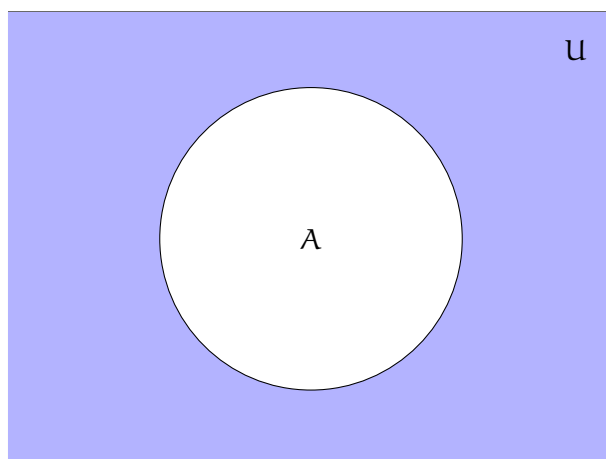
$$A^c = U - A$$

$$A^c = \{x \in U \mid x \notin A\}$$

$$U = \{1, 2, 3, 4, 5\}, \quad A = \{1, 2, 3\}$$

$$A^c = \{4, 5\}$$

$\therefore A^c$ is the complement of A.



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Información para evaluadores:

Cartesian product

The Cartesian product of two sets A and B is the set of all ordered pairs (a,b) , where a is an element of A and b is an element of B .

Symbol: $A \times B$

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

if number of elements in A is m and B is n then,

$$|A \times B| = |A||B| = mn$$

$$A = \{1, 2\}, \quad B = \{a, b\}$$

$$A \times B = \{(1, a), (1, b), (2, a), (2, b)\}$$

$\therefore A \times B$ is the Cartesian product of A and B .

10 Set builder form and roster form

No.	Set-builder Form	Roster Form
1	$A = \{x \in \mathbb{N} \mid x < 6\}$	$A = \{1, 2, 3, 4, 5\}$
2	$B = \{x \in \mathbb{N} \mid x \text{ is even, } x < 10\}$	$B = \{2, 4, 6, 8\}$
3	$C = \{x \in \mathbb{N} \mid x \text{ is odd, } x < 10\}$	$C = \{1, 3, 5, 7, 9\}$
4	$D = \{x \in \mathbb{Z} \mid -2 \leq x \leq 2\}$	$D = \{-2, -1, 0, 1, 2\}$
5	$E = \{x \in \mathbb{N} \mid x \text{ is prime, } x < 15\}$	$E = \{2, 3, 5, 7, 11, 13\}$
6	$F = \{x \in \mathbb{N} \mid x \text{ is a multiple of 3, } x < 15\}$	$F = \{3, 6, 9, 12\}$
7	$G = \{x \in \mathbb{N} \mid x^2 < 25\}$	$G = \{1, 2, 3, 4\}$
8	$H = \{x \in \mathbb{Z} \mid x^2 = 9\}$	$H = \{-3, 3\}$
9	$I = \{x \in \mathbb{N} \mid 2 < x \leq 7\}$	$I = \{3, 4, 5, 6, 7\}$
10	$J = \{x \in \mathbb{N} \mid x \text{ is a factor of 12}\}$	$J = \{1, 2, 3, 4, 6, 12\}$

Hasse diagram

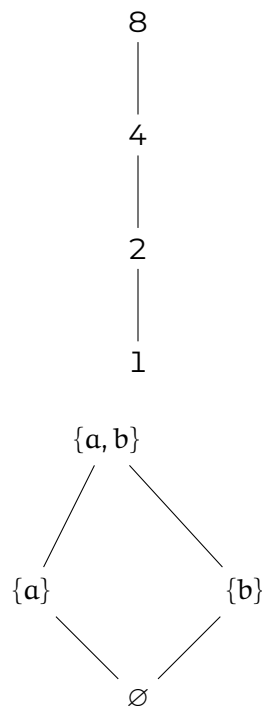
A Hasse diagram is a graphical representation of a partially ordered set (poset). It shows the ordering relationships between elements while omitting reflexive and transitive relationships.

For example, consider:

$$A = \{1, 2, 4, 8\}$$

$$B = \{a, b\}$$

The Hasse diagram is:



Principle of inclusion and exclusion

The Principle of Inclusion and Exclusion is a counting technique used to find the number of elements in the union of sets, especially when the sets overlap.

Principle of Inclusion and Exclusion

Two Sets

For two finite sets A and B,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Three Sets

For three finite sets A, B, and C,

$$\begin{aligned}
 |A \cup B \cup C| &= |A| + |B| + |C| \\
 &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\
 &\quad + |A \cap B \cap C|
 \end{aligned}$$

General Case

For n finite sets $A_1, A_2, \dots, A_n$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

Two Sets

Suppose in a class of 50 students, 30 students like Mathematics, 25 students like Physics, and 10 students like both. Using the Principle of Inclusion and Exclusion,

$$|M \cup P| = |M| + |P| - |M \cap P|$$

$$= 30 + 25 - 10$$

$$|M \cup P| = 45$$

Therefore, 45 students like at least one of the two subjects.

Three Sets

In a class, 40 students like Mathematics, 30 like Physics, and 20 like Chemistry. Also,

$$|M \cap P| = 10, \quad |M \cap C| = 8, \quad |P \cap C| = 5$$

and

$$|M \cap P \cap C| = 3.$$

By the Principle of Inclusion and Exclusion,

$$\begin{aligned}
 |M \cup P \cup C| &= |M| + |P| + |C| \\
 &\quad - |M \cap P| - |M \cap C| - |P \cap C| \\
 &\quad + |M \cap P \cap C|
 \end{aligned}$$

$$= 40 + 30 + 20 - 10 - 8 - 5 + 3$$

$$|M \cup P \cup C| = 70$$

Therefore, 70 students like at least one of the three subjects.

Partially ordered set (Poset)

A partially ordered set (poset) is a set together with a relation that is reflexive, antisymmetric, and transitive. Let $A = \{1, 2, 3, 6\}$ and define the relation by divisibility:

Let

$$A = \{1, 2, 3, 6\}$$

and define the relation $\leq$ by divisibility:

$$a \leq b \iff a \mid b$$

Then,

$$1 \mid 2, \quad 1 \mid 3, \quad 1 \mid 6, \quad 2 \mid 6, \quad 3 \mid 6.$$

Thus, $(A, \mid)$ is a partially ordered set (poset).